

Scalable Parallel-in-Time Integration for Equations of Motion

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Table of Contents

Introduction

Background

The Parareal Algorithm

The Parareal Algorithm at Scale

Performance Analysis

Conclusion

Content

Introduction

Background

The Parareal Algorithm

The Parareal Algorithm at Scale

Performance Analysis

Conclusion

The Main Issue

Calculations take time...

The Main Issue

Calculations take time...and we only have so much.

The Future is Parallel

- Total runtime = $\frac{\text{time}}{\text{calculation}} * \text{number of calculations}$
- Total runtime $\downarrow \iff \frac{\text{time}}{\text{calculation}} \downarrow$ or $\frac{\text{calculation}}{\text{time}} \uparrow$

The Future is Parallel

- Total runtime = $\frac{\text{time}}{\text{calculation}} * \text{number of calculations}$
- Total runtime $\downarrow \iff \underbrace{\frac{\text{time}}{\text{calculation}}}_{\text{Sequential}} \downarrow \text{ or } \underbrace{\frac{\text{calculation}}{\text{time}}}_{\text{Parallel}} \uparrow$

The Future is Parallel

- Total runtime = $\frac{\text{time}}{\text{calculation}} * \text{number of calculations}$
- Total runtime $\downarrow \iff \underbrace{\frac{\text{time}}{\text{calculation}}}_{\text{Sequential}} \downarrow \text{ or } \underbrace{\frac{\text{calculations}}{\text{time}}}_{\text{Parallel}} \uparrow$
- Hardware improvements $\Rightarrow \frac{\text{time}}{\text{calculation}} \downarrow$, but it's decelerating
- More hardware $\Rightarrow \frac{\text{calculation}}{\text{time}} \uparrow$, and it's accelerating

The Future is Parallel

Parallel algorithms need to be the focus

Content

Introduction

Background

The Parareal Algorithm

The Parareal Algorithm at Scale

Performance Analysis

Conclusion

Content

Introduction

Background

The Parareal Algorithm

The Parareal Algorithm at Scale

Performance Analysis

Conclusion

Review of Equations of Motion

Content

Introduction

Background

The Parareal Algorithm

The Parareal Algorithm at Scale

Performance Analysis

Conclusion

Review of High-Performance Computing

Parareal on the GPU

Content

Introduction

Background

The Parareal Algorithm

The Parareal Algorithm at Scale

Performance Analysis

Conclusion

Numerical Analysis

Latency & Data Transfers

Content

Introduction

Background

The Parareal Algorithm

The Parareal Algorithm at Scale

Performance Analysis

Conclusion

Thanks for Listening.